

HANOI UNIVERSITY OF
SCIENCE AND TECHNOLOGY

SCHOOL OF INFORMATION AND COMMUNICATION
TECHNOLOGY

DATA VISUALIZATION REPORT

Hong Kong Bus Network Analytics Dashboard

Course: Data Visualization

Class: 168068

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Abstract

This project developed an interactive Hong Kong Bus Network Analytics Dashboard designed to support Urban Planners and Transport Policy Analysts in monitoring transit infrastructure and service equity. Sourced from the official DATA.GOV.HK "Routes and Fares of Public Transport" dataset for the 2024–2026 period, the dashboard transforms technical logistical tables into a coordinated analytical story.

The dashboard follows a top-down narrative flow, guiding users from territory-wide Network KPIs to operator benchmarking and detailed spatial distribution. Developed using Python (Streamlit, Plotly, and Pydeck), the application strictly adheres to core visualization principles.

By integrating temporal growth timelines, interactive flow maps, and distributional diagnostics, the tool enables analysts to identify network anomalies and pricing outliers. Ultimately, the dashboard demonstrates how applying perception-based design and coordinated multi-panel layouts can turn granular transit records into a powerful lens for infrastructure management and passenger equity analysis.

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1 Dataset overview

This project uses publicly available data on Hong Kong’s public transport network, specifically the “Routes and Fares of Public Transport” dataset published by the Hong Kong Transport Department through the Hong Kong Open Data Portal.

The dataset provides official records of public transport routes, operators, service modes, and fare structures across Hong Kong. It is maintained as part of the government’s open data initiative and is intended for public use, research, and analysis.

The data was accessed via the following sources:

- Hong Kong Open Data Portal (Transport Department dataset page): https://data.gov.hk/en-data/dataset/hk-td-tis_3-routes-and-fares-of-public-transport
- Google Dataset Search entry for the same dataset: <https://datasetsearch.research.google.com/search?src=0&query=hong%20kong%20routes%20and%20fares&docid=L2cvMTF0eG5xOHM4ag%3D%3D>

The dataset covers the period from 1 June 2024 to 30 May 2026, enabling analysis of recent and up-to-date changes in Hong Kong’s public transport system.

1.1 Dataset Structure and Schema

Building on the data source, this section provides a structured overview of the dataset used in this project. It introduces the underlying relational schema and describes the main tables that compose the database, including their key fields, data types, and meanings. Together with the entity-relationship diagram (Figure 1, this section explains how route, stop, fare, and operator information are organized and linked, forming the foundation for subsequent data processing and analysis.

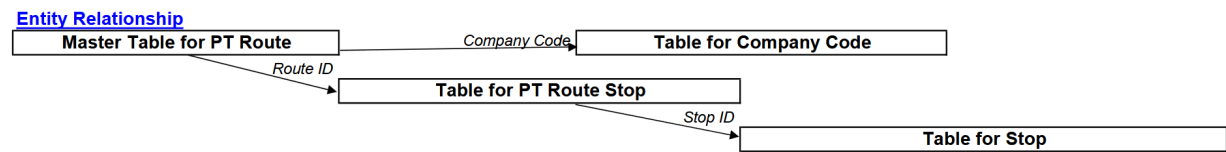


Figure 1: Entity-Relationship Diagram

Table 1: Schema for Master Table for PT Route

Field	Data Type	Description
ROUTE_ID	Integer	Route ID
COMPANY_CODE	String	Operator code (e.g., KMB, CTB)
DISTRICT	String	HKI, KLN, NT (GMB only)
ROUTE_NAMEC/S/E	String	Route Name (Chinese/Simplified/English)
ROUTE_TYPE	Integer	1=bus, 2=GMB, 4=tram, 5=peak tram, 7=ferry
SERVICE_MODE	String	A=whole, N=night, R=regular, etc.
SPECIAL_TYPE	Integer	0=N/A, 1=Time specific, 2=Weekend fare, 3=Both
JOURNEY_TIME	Integer	Journey Time in minutes
FULL_FARE	Numeric	Full fare amount
LOC_START_NAMEC/S/E	String	Start Location (Chinese/Simplified/English)
LOC_END_NAMEC/S/E	String	End Location (Chinese/Simplified/English)
LAST_UPDATE_DATE	Datetime	Date and time of last update

Table 2: Schema for Table for PT Route Stop

Field	Data Type	Description
ROUTE_ID	Integer	Route ID
ROUTE_SEQ	Integer	1 = outbound or circular route; 2 = inbound
STOP_SEQ	Integer	Stop sequence of a route
STOP_ID	Integer	Unique Stop ID
STOP_NAMEC/S/E	String	Stop name (Chinese/Simplified/English)
LAST_UPDATE_DATE	Datetime	Date and time of last update

Table 3: Schema for Table for Stop

Field	Data Type	Description
STOP_ID	Integer	Unique Stop ID
STOP_TYPE	Integer	1=bus, 2=GMB, 5=tram, 6=peak tram, 8=ferry
X	Integer	x-coordinate (Hong Kong 1980 Grid system)
Y	Integer	y-coordinate (Hong Kong 1980 Grid system)
LAST_UPDATE_DATE	Datetime	Date and time of last update

Table 4: Schema for Table for Fare

Field	Data Type	Description
ROUTE_ID	Integer	Route ID
ROUTE_SEQ	Integer	1 = inbound or circular; 2 = outbound
DAY_CODE	Integer	Code for applicable days (e.g., 511 = All Days)
ON_SEQ	Integer	Stop sequence of pick-up stop
OFF_SEQ	Integer	Stop sequence of drop-off stop
PRICE	Numeric	Fare from pick-up to drop-off
LAST_UPDATE_DATE	Datetime	Date and time of last update

Table 5: Schema for Table for Company Code

Field	Data Type	Description
COMPANY_CODE	String	Company / Operator code
COMPANY_NAMEC/S/E	String	Operator name (Chinese/Simplified/English)
DESCRIPTION	String	Additional description of company code

1.2 Derived Features

In addition to the raw dataset provided by the Hong Kong Transport Department, several derived features were engineered to support network analysis, temporal comparison, and visualization of the public transport system. These features were computed from the base tables and are not explicitly present in the original dataset.

The derived variables can be grouped into four main categories: stop-level features, route-level features, temporal change features, and network edge features.

1.2.1 Stop-Level Derived Features

Stop-level features capture the structural importance and geographic representation of each stop in the transport network.

Table 6: Derived Features for Stops

Feature	Description
lon	Longitude converted from Hong Kong 1980 grid coordinates
lat	Latitude converted from Hong Kong 1980 grid coordinates
degree	Number of routes passing through a stop, representing connectivity

The degree feature is particularly important as it represents the centrality of a stop within the transport network, allowing identification of major interchange hubs.

1.2.2 Route-Level Derived Features

Route-level features describe operational characteristics of each public transport route and enhance comparative analysis across operators.

Table 7: Derived Features for Routes

Feature	Description
COMPANY_NAME	Expanded operator name mapped from COMPANY_CODE
PATH	Geospatial representation of the route for mapping visualization

The PATH feature is used to construct polyline geometries for visualizing routes on the network map.

1.2.3 Temporal Change Features

Temporal features are derived from snapshot-based comparisons of the network over time. These allow tracking of how the transport system evolves.

Table 8: Derived Features for Temporal Changes

Feature	Description
stop_changes	Number of stops added, removed, or modified between snapshots
route_changes	Changes in total route count over time

These features enable trend analysis of network expansion and contraction over time.

1.2.4 Network Edge Features

Edge-level features represent relationships between stops and capture the structure of the transport network as a graph.

Table 9: Derived Features for Network Edges

Feature	Description
from_stop / to_stop	Start and end stop identifiers for each connection
from_stop_coords / to_stop_coords	Geographic coordinates of connected stops
routes	List of routes passing through the edge
route_count	Number of routes sharing the same edge
companies	List of operators servicing the edge
company_count	Number of unique operators on the edge

These features form the basis of the network graph representation used in spatial visualisation and connectivity analysis.

2 Dashboard Overview

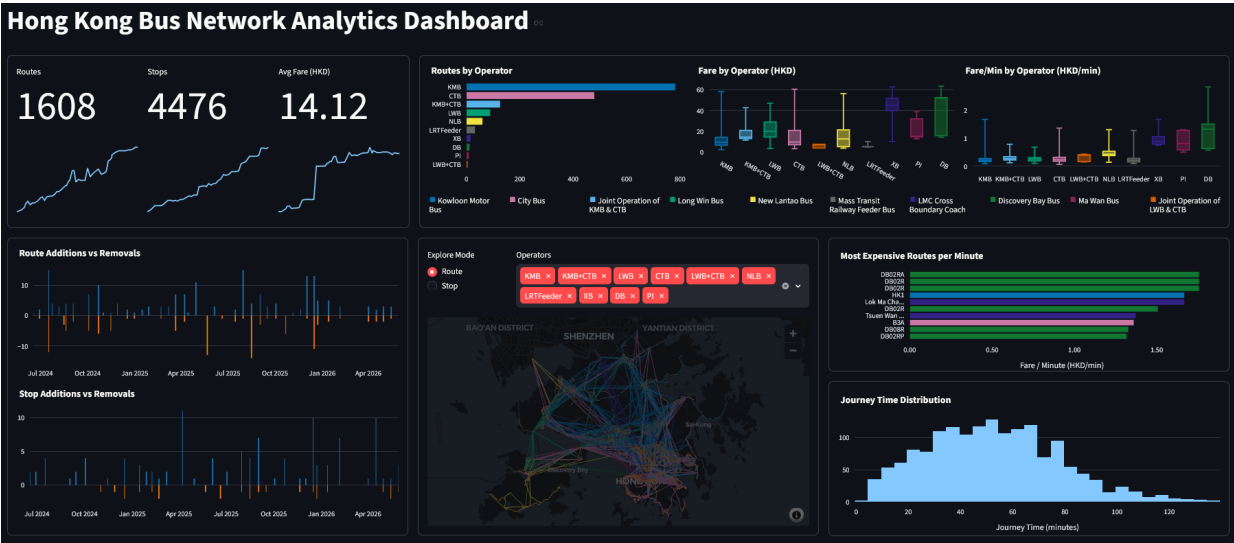


Figure 2: Hong Kong Bus Network Analysis Dashboard

2.1 Scenario and Target Audience

The Hong Kong Bus Network Analytics Dashboard is designed as a monitoring tool for urban planners and transport policy analysts. This system is intended to support rapid interpretation of high-level strategic questions, including:

- **Network Health:** Is the overall transport infrastructure (routes and stops) expanding or contracting over time?
- **Operator Performance:** How do major companies compare in terms of network share and pricing?

- Economic Efficiency: Which routes represent outliers with disproportionately high fares per minute?

2.2 Dashboard Design Logic

The dashboard is structured as a coordinated multi-panel layout, following a top-down analytical narrative that progresses from aggregate summaries to detailed spatial and operational insights:

- Top Row (Summary & Comparison): Establishes a system-wide baseline using KPI indicators and operator benchmarks.
- Bottom Left (Temporal Dynamics): Visualises net changes in the transport network over the observation period.
- Center (Spatial Exploration): Provides a geographic representation of the network, enabling spatial interpretation of connectivity.
- Bottom Right (Distributions & Anomalies): Highlights service characteristics and identifies economic outliers.

2.3 Main Dashboard Sections

- Network Snapshot (Top Left): Three primary KPI cards display total routes, total stops, and average fare.



Figure 3: KPI cards with sparklines

- Metric Status: Large numerical values provide immediate system-level indicators.
- Trend Context: Embedded sparklines show historical evolution from 2024 to 2026 without requiring additional charts.
- Operator Benchmarking (Top Right):

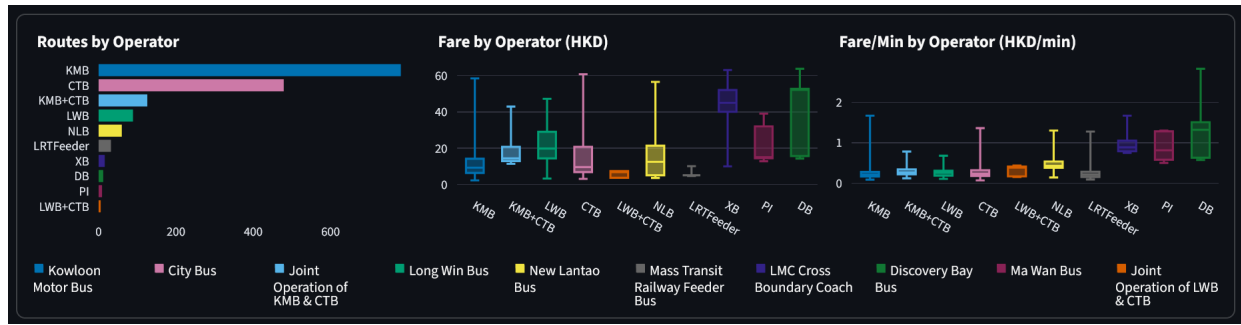


Figure 4: Operator-related charts

- Routes by Operator: A horizontal bar chart ranking operators by total route count, enabling quick identification of dominant providers.
- Fare & Efficiency Box Plots: Distributional views of fares and fare-per-minute metrics, highlighting variability and pricing structure differences across operators.
- Infrastructure Growth Timeline (Bottom Left): Paired bar charts showing route and stop additions versus removals over time.

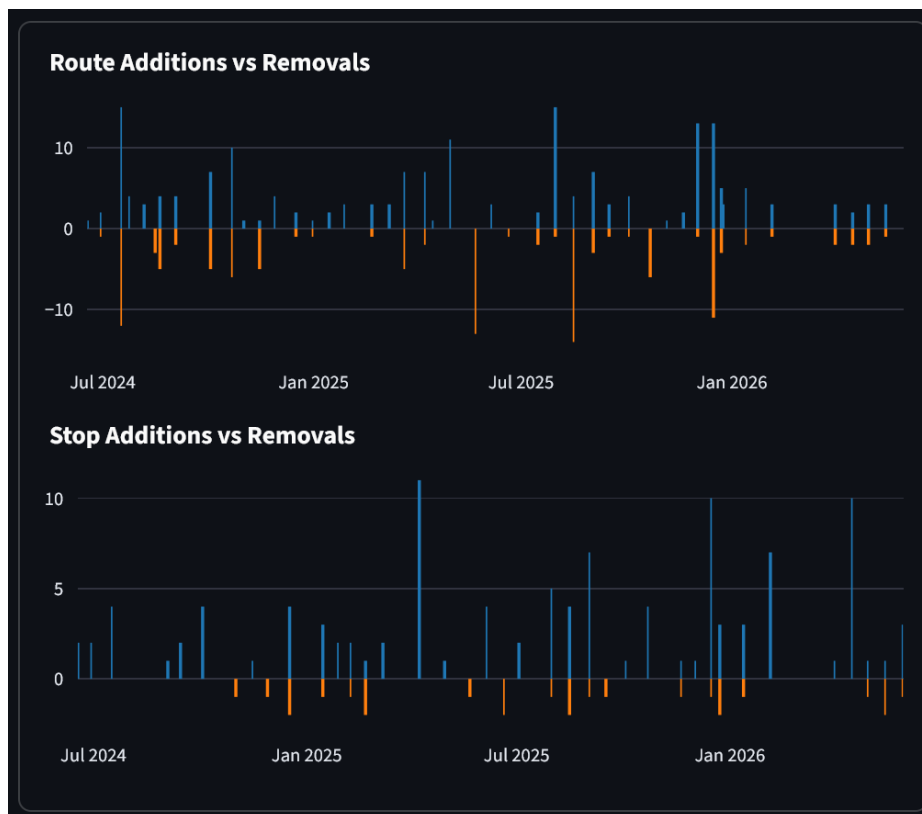


Figure 5: Growth charts

- Design Rationale: Blue (additions) and orange (removals) are anchored on a shared zero baseline, enabling clear interpretation of net system growth or decline.
- Spatial Network Map (Center): An interactive PyDeck-based map visualising the connectivity of the bus network.

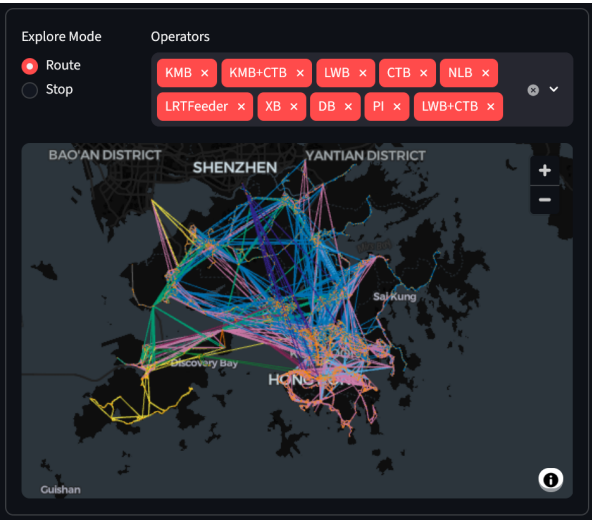


Figure 6: Route Map

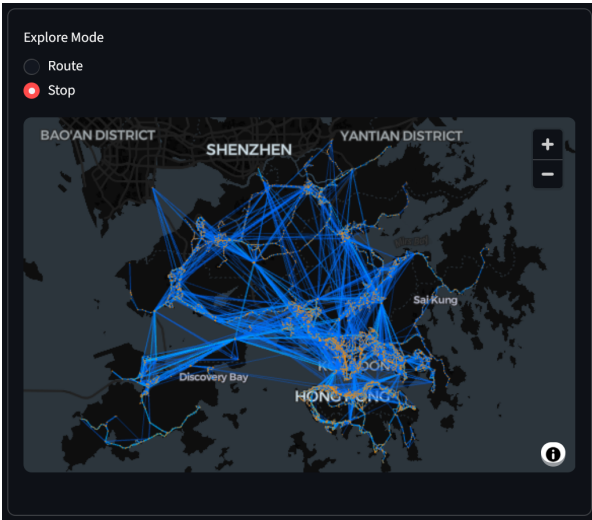


Figure 7: Stop Map

- Interaction: Users can switch between “Route” and “Stop” modes and filter by operator to isolate specific network segments.
- Service Efficiency & Anomalies (Bottom Right):



Figure 8: Journey Time Distribution & Top 10 most Expensive Routes

- Journey Time Distribution: A histogram representing the distribution of trip durations across the network.
- Most Expensive Routes per Minute: A ranked horizontal bar chart identifying routes with the highest fare-per-minute values (e.g., DB02RA), highlighting low value-for-money services.

2.4 Rationale for Visual Choices

- **Consistent Categorical Colouring:** A fixed COMPANY_COLORS mapping is used throughout the dashboard. Operators retain consistent colours across all charts and map layers, supporting pre-attentive visual grouping.
- **Visual Integrity:** A colorblind-safe palette (Okabe & Ito) is used consistently across all charts to ensure perceptual accessibility without compromising categorical distinction.
- **Proximity and Closure:** Following Gestalt principles, related visualisations (e.g., fare distribution box plots) are grouped to encourage joint interpretation as a single analytical unit.
- **Maximised Data-Ink Ratio:** A minimal design approach is used with reduced chart clutter, direct labeling, and selective removal of redundant legends to improve readability.

3 Technique Application

This section analyzes the visualization techniques and principles from the course applied to the Hong Kong Bus Network Analytics Dashboard.

3.1 Chapter 1: Introduction — Why Visualize, and What Kind

Techniques / Principles Applied

- **Motivation:** Using visualization to amplify cognition and uncover patterns in complex transit data.
- **Role of Visualization:** Moving from exploratory analysis (discovery) to explanatory communication (sharing insights).
- **Visualization Types:** Adopting a hybrid approach that supports both high-level summaries and detailed exploration.

Application in the Dashboard

The dashboard is designed for urban planners to monitor the health and efficiency of Hong Kong's bus network. It addresses the challenge of data density by transforming raw route tables into an interpretable visual narrative.

- **Exploratory:** Users can filter by operator or zoom into the geographic map to identify route clusters and spatial gaps.
- **Explanatory:** KPI cards provide an immediate system status summary, including total routes (1,608), stops (4,476), and average fare (\$14.12).

3.2 Chapter 2: Data Types, Marks, and Channels

Techniques / Principles Applied

- **Data types:** Nominal (N), Ordinal (O), and Quantitative (Q) variables.
- **Visual marks:** points, lines, and areas.
- **Visual channels:** position, length, and color based on perceptual effectiveness.

Application in the Dashboard

- Data Types:
 - Nominal: Operator (COMPANY_CODE), Route ID.
 - Quantitative: FULL_FARE, JOURNEY_TIME, route counts.
- Marks and Channels:
 - Color (Nominal): Operators are encoded using consistent categorical colors across all views.
 - Position (Quantitative): Used in box plots and histograms for accurate comparison of distributions.
 - Length (Quantitative): Used in bar charts such as “Routes by Operator” for precise ranking.
- Rationale: Following Mackinlay’s ranking, position and length were prioritized for quantitative accuracy over less precise visual channels.

3.3 Chapter 3: Pre-attentive Processing, Gestalt, and Magnitude Estimation

Techniques / Principles Applied

- Pre-attentive processing (visual pop-out effects)
- Gestalt principles (proximity, similarity, connectedness)
- Magnitude estimation and perceptual accuracy

Application in the Dashboard

- Pre-attentive Processing: Consistent operator colors allow immediate visual identification across all panels.
- Gestalt Principles:
 - Proximity: KPI cards are grouped to represent system-wide summary statistics.
 - Similarity: Consistent color mapping creates cross-panel associations.
 - Connectedness: Sparklines and network edges visually connect related data points.
- Magnitude Estimation: Bar charts are preferred over area-based encodings to avoid perceptual distortion.

3.4 Chapter 4: Chart Types for Amounts, Distributions, and Trends

Techniques / Principles Applied

- Task-driven visualization design
- Coordinate systems (Cartesian and geographic)

Application in the Dashboard

- Amounts: Bar charts represent route counts and identify high-cost routes.
- Distributions: Box plots and histograms show variability in fares and journey times.
- Trends: Sparklines summarize temporal changes from 2024–2026.
- Growth Dynamics: Diverging bar charts show additions and removals of routes and stops over time.

3.5 Chapter 5: Graph Data

Techniques / Principles Applied

- Node-link graph representation
- Complexity reduction through filtering

Application in the Dashboard

- Stops are modeled as nodes, and connections between stops as edges.
- Filters reduce edge density and prevent overplotting in dense network regions.

3.6 Chapter 6: Visual Integrity and Design

Techniques / Principles Applied

- Proportional ink principle
- Data-ink ratio maximization
- Coordinated multi-panel design
- Color palette accessibility

Application in the Dashboard

- 3D effects are avoided to prevent perceptual distortion (Lie Factor reduction).
- A coordinated layout ensures synchronized filtering across all panels.
- A qualitative palette was chosen because its colors are designed to be distinguishable from each other under all forms of color-vision deficiency.

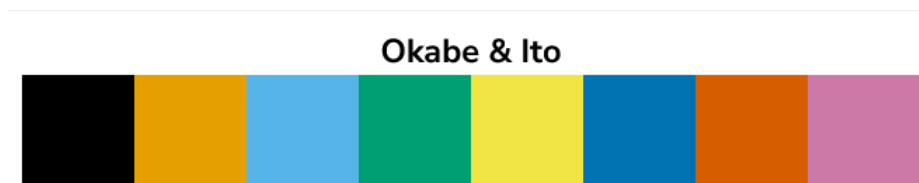


Figure 9: Okabe & Ito color palette

3.7 Chapter 7: Map Visualization

Techniques / Principles Applied

- Geographic visualization types (flow, point, and network maps)
- Geocoding and spatial projection

Application in the Dashboard

- A flow-based network map visualizes connectivity across Hong Kong.
- Color encodes operator identity, while line density reflects network concentration.
- Stop coordinates from the Hong Kong 1980 Grid system are converted for spatial plotting.

3.8 Chapter 8: Interaction

Techniques / Principles Applied

- Filtering, zooming, and details-on-demand
- Interactive visualization tools

Application in the Dashboard

- Global operator filters enable cross-panel updates.
- Mode switching allows toggling between route-level and stop-level views.
- Tooltips provide detailed values on hover for precise interpretation.

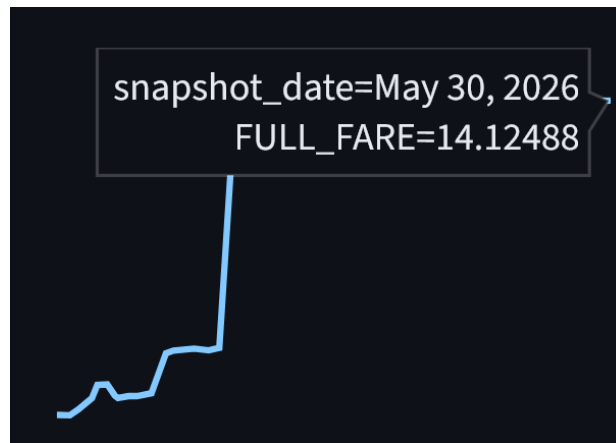


Figure 10: Tooltip

3.9 Chapter 9: Storytelling

Techniques / Principles Applied

- Narrative visualization structure (beginning, middle, end)
- Balance between exploratory and explanatory design

Application in the Dashboard

- Beginning: KPI cards provide a high-level system overview.
- Middle: Operator benchmarking and temporal trends provide analytical depth.
- End: Outlier detection highlights inefficient or high-cost routes for policy attention.

4 Limitations and Summary

4.1 Limitations

Despite the application of course principles to ensure visual integrity and effectiveness, several technical and theoretical limitations remain in the current dashboard:

- **Geographic Rendering Performance:** The interactive PyDeck map, which visualizes nearly 4,476 stops and over 1,600 routes, experiences slow loading times when rendering the full network. While caching was implemented for data loading, the client-side rendering of thousands of complex path segments remains a performance bottleneck.
- **Interaction Encoding Constraints:** A key limitation of the Streamlit framework is that the operator selection widget (multi-select) does not support visual encoding. According to the Principle of Consistency, selection labels should ideally include color-coded indicators aligned with the `COMPANY_COLORS` mapping to support pre-attentive processing.
- **Categorical Complexity:** Although a colorblind-friendly palette (Okabe & Ito) was used to improve accessibility, the dataset contains 10 distinct operator categories. Course theory suggests that perceptual discrimination of categorical hues becomes less reliable as the number of categories increases, potentially reducing clarity when all operators are displayed simultaneously.
- **Occlusion and Clutter:** In Route mode, the density of the Hong Kong transit network leads to significant overplotting. Although transparency is used for route rendering, the central network area still forms a dense network, making individual routes difficult to distinguish without filtering.

4.2 Summary

This project successfully developed an interactive dashboard for analyzing the Hong Kong bus network using data from 2024 to 2026. By transforming raw structured datasets from the Hong Kong Open Data Portal into a coordinated visual analytics interface, the dashboard enables urban planners to explore network growth, operator performance, and service efficiency.

The main takeaway from this work is that effective dashboard design requires a coherent top-down narrative structure. The system guides users from high-level network KPIs, to operator benchmarking, and finally to spatial and distributional insights, culminating in the identification of anomalies such as the “Most Expensive Routes per Minute.” This ensures that insights are interpreted in context rather than in isolation.

Ultimately, the application of visualization principles such as Mackinlay’s channel ranking, Gestalt grouping, and colorblind-friendly design enhances both interpretability and accessibility. The dashboard demonstrates how transit infrastructure data can be transformed into an effective analytical tool for monitoring system health and supporting evidence-based transport planning.